

Towards Object-oriented Storytelling via 1st and 3rd Person Perspectives

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Figure 1: Using storytelling in family education for comity and respect: *while two goats meeting at a single-plank bridge.* (a) The 1st person perspective of the brown goat: The brown goat sees a grey goat standing at the other side of the single-plank bridge from its own perspective. (b) This is the 3rd person perspective of the two goats. Parents and children can imagine what they should do or how they can help the two goats. The parents can take the role as story director. If the two goats do not yield way, they cannot pass the bridge. (c) This is the 1st person perspective of the grey goat. The parents can use the first person perspective to teach their children to yield way to others. The storytelling system can simulate the outcomes of the decision taken by a goat. Therefore, the parents can teach the children about comity and respect.

ABSTRACT

Joint media engagement (JME) provides digital interaction and becomes a convenient tool for collaborative storytelling. However, the person perspectives, reusability, and modifiability of the story contents are rarely discussed; also parents may lack the skills of professional story-making. Therefore, we propose an object-oriented three-dimensional (3D) storytelling system, which provides an interactive educational media space for parent-child storytelling. The system supports manipulating reusable character objects and narrating stories from the 1st and 3rd person perspectives in an intuitive manner. Parents can create stories freely based on the background of famous fairy tales and incorporate family education. Our goal is to create a storytelling system for educational entertainment that is fun and conducive to free expression and creativity.

CCS CONCEPTS

• Human-centered computing; • Interaction design;

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KEYWORDS

Family education, object-oriented, interactive storytelling, collaborative JME, perspective-taking

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1 INTRODUCTION

In children growth, family education is the most important stage in education. It greatly influences the majority of children life habits, learning attitudes, different disposition characteristic, and life values in the future [1]. It is also related to the harmonious development of children abilities in language, cognition, emotion, physical strength, social skills [2]. Chen defined family education as *a society used to educate new generations with aims and systems* [3]. Dewey proposed the center of children theory that children are the center of education [4]. Particularly, the communication between parents and children, which conveys all the messages of mutual trust between each other. The virtuous parent-child interactions can create a subsequent good family atmosphere. Therefore, family education is usually accompanied by the interactive behaviors of parent-child discussion. Communication fosters the human feelings of closeness [5] and has a good learning effect on family education.

A recent study has reported that the experiences of parent-child interaction contribute to cognitive, emotional, and social wellbeing in childhood [6]. Storytelling can establish an extensive oral language base for children development [7]. Therefore, it has become one of the productive parent-child interaction activities for children. However, most parents usually use books that have visual illustrations along with text to convey the information and meaning of stories to their children. Although the stories of the books with visual appealing contents have been created by experts, the story contents cannot be modified or enhanced by parents interactively [8]. Furthermore, the outcome of the story is also not necessarily the education that parents want to give their children, such as the girl taking revenge on the big wild wolf at the ending of "Little Red Riding Hood". As there are increasingly interactive storytelling media [9], JME makes ease, reliability, and diversity for storytelling [10-12]. Using JME, parents can create family stories themselves or with their children together. However, parents may lack the skills of professional story-making to produce compelling story contents for attracting the attention of their children. Therefore, we would like to develop a storytelling system which provides an interactive media space that allows parents to tell stories to their children in an interactive manner. The parents can modify the behaviors of the main story characters so that they can incorporate family education.

In this paper, we propose a novel interaction storytelling system, which supports the creation of family stories in a 3D virtual environment via narrating stories with real-time character operations. The system allows the story content to be adjusted in a dynamic manner. It enables parents and children to participate in the process of storytelling to achieve a rich and vivid media space that is educational entertainment. We adopt the object-oriented concept so that family members can manipulate story characters in an intuitive manner. Furthermore, the system allows users to switch seamlessly between the 1st person and 3rd person perspectives. Parents can choose to tell stories from the first-person perspective of a certain story character or switch to the third-person perspective to watch the story progress during storytelling. Using different person perspectives is conducive to parents teaching family education to their children when the concerns of each main character or all the characters are considered. The first person perspective is good for perspective-taking while the third person perspective allows people to think deeply about the relationship of the main story characters and how the characters affect each other. The parents can guide their children to understand well the outcomes under these two person perspectives. Figure 1 presents the system application of storytelling.

2 RELATED WORK

Storytelling activities help children develop emotional [13], oral language base [7, 14], social experiences [15], interpersonal and communication skills [16]. Children can make sense of the world by creating story characters and telling stories. As computers, smartphones, and tablets advance, they are employed in various creative applications for storytelling [17-19]. Especially, the collaborative storytelling mechanism promotes opportunities for user interaction. Hourcade *et al.* showed *KidPad*, which was a collaborative

storytelling tool for children, could provide children with interaction interfaces in a collaborative environment [16, 20]. Russell implemented a storytelling and animation tool, *ToonTastic*, which allowed children to create and share their own stories with others [21]. Quinn *et al.* developed an iPhone application, *StoryKit*, to create and share audio-visual stories. It supported children to make stories with elders together and could let users create books on the touchscreen device via arranging the text, photos, and sounds [22]. Fails *et al.* presented *Mobile Stories*, a mobile technology, which empowered children to collaboratively read and create stories [23-26]. Fono *et al.* developed an application of mobile phones, *Sandboxes*, that offered a collaborative multimedia composition [27]. It was interactivity, flexibility, and cohesiveness. The following summarizes the features of collaboration activities in storytelling [28-29]:

Users create their own story materials that can be collected to create a synthesized artifact for storytelling.

Each story material, which is a story unit, is fixed after being integrated into a story; it can be stored and reused for storytelling.

Users collaborate using JME one after the other or at the same time to create the story content, e.g., painting or diary.

Most research of collaboration stories focuses on the interaction benefits of digital storytelling co-authoring tasks [22, 30-35]. However, the reusability of the results of collaborative story creation is rarely discussed. On the other hand, the creators of collaborative stories are sometimes only assigned to produce the part of the story. Therefore, most creators only focus on how to complete assigned tasks and might not be able to have enough chance or motivation to understand the story as a whole [28]. Furthermore, the integration and uploading of story materials is also a challenge for children [28]. We considered that the contents of stories were important and precious for interesting storytelling. Reusing existing story contents was useful for extending the children story experience. Therefore, we proposed the concept of object-oriented storytelling for using the character objects as the story contents [36]. All story contents have their features and can be reused during storytelling, including the story characters and scenes. Users can manipulate character objects in an intuition approach or change their appearance to create a dynamic story world. In this work, we further develop an object-oriented 3D storytelling system, using the characteristics of objects that can be reused or reproduced to create a virtual story scene. Furthermore, as asymmetry of digital technology is defined as the capacity of team members to have different ways to visualize and interact with virtual content [38], the asymmetries of information were used to develop communication and coordination between players [39]. In particular, operation connections between different perspectives can facilitate collaboration or interaction [44-46]. We employ the concept of asymmetric perspective [37] which enables parents to stand in the position of different character roles to tell a story. We believe the proposed system can provide a family an interactive media space to create a plentiful story environment that facilitates parents to educate their children by storytelling from different person perspectives.

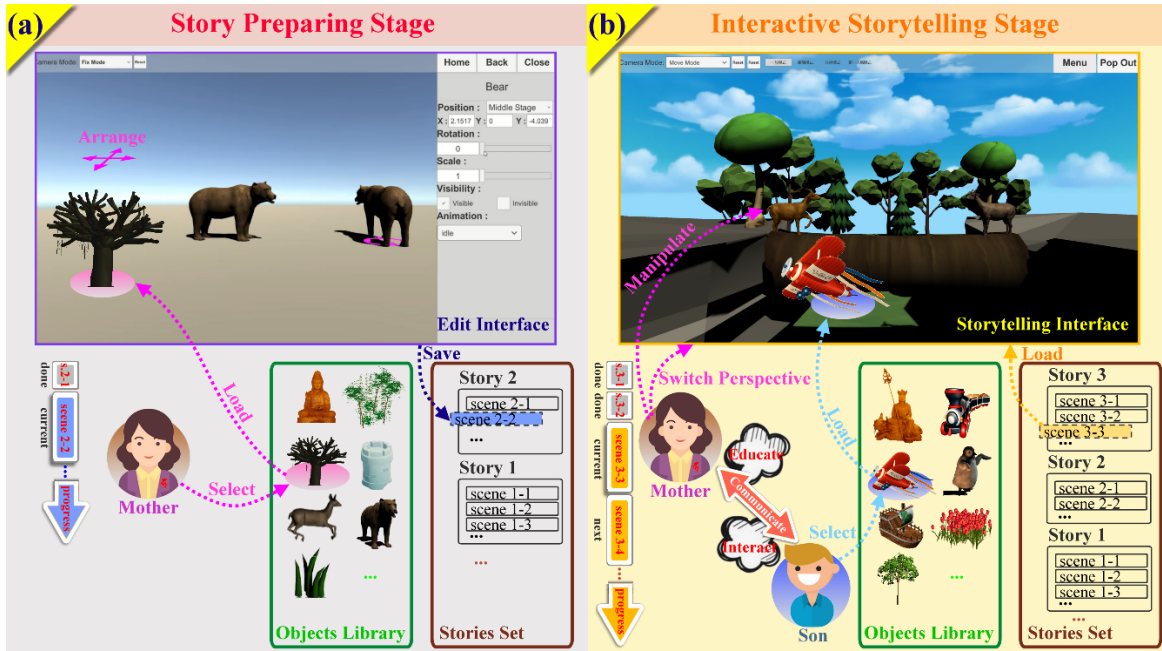


Figure 2: System overview and storytelling activities of a family. (a) A mother creates the story scenes before storytelling. (b) Her child joins the storytelling in a virtual 3D story playground. The mother narrates the story and interactively manipulates the story characters to achieve a vivid educational media space.

3 SYSTEM DEVELOPMENT

In this section, we describe the design and implementation of the proposed storytelling system, which provides an interactive media space for families to participate simultaneously in story performance. It enables parents and their children to interactively manipulate character objects. We build the system based on the object-oriented concept [36] and asymmetric play design in HCI [40–43]. Figure 2 presents the system overview and storytelling activities of users.

3.1 System architecture

We implement the storytelling system on Unity which supports character animation and physics simulation in virtual environments. The system consists of three major components: an object manager, a story manager, and a graphical user interface. The following describes the functions of the components:

The object manager manages an object library that contains story materials. The story materials are categorized into groups (e.g., animals, plants, and scene models) for ease management. Users can retrieve the required items from the library for story preparation or activate them during storytelling.

The story manager manages a set of stories. The stories are created by users. Each story has its own story scenes in which there are story characters and objects.

The graphical user interface provides a virtual room for users to manipulate the story characters and objects during the story preparation stage or storytelling stage.

We implemented the storytelling system on Microsoft Visual Studio 2017 (C# software development) using Unity 3D 2018.3.12f1

(storytelling engine). The laptop contained the Intel® Core™ i7-6700 CPU, NVIDIA GeForce GTX 1080 with RAM 16.0 GB.

3.2 Story contents

In general, a story is composed of a series of scenes and each scene has some story characters and objects. The story characters are human characters and animals. The objects include trees, grasses, rivers, rocks, clouds, sun, and so on. The story characters are three-dimensional models which are associated with skeletons. Thus, they can be animated to perform different actions (e.g., chatting, walking, running, fighting). In each scene, the listeners and the storyteller manipulate the story characters to make the characters interact with each other. In our stories, two characters interact with each other in most scenes. The interaction actions of two characters include *talking*, *walking together*, *running together*, *approaching each other*, and *fighting with each other*. Furthermore, the characters may have a relationship, such as *friend*, *enemy*, and *stranger*. To trigger an interaction action, the users select the two characters. Then the interaction actions of the two characters are displayed around the characters so that the users can trigger an action. To make users intuitively manipulate the characters, we adopt a context-aware approach to display interaction actions for characters. The main idea is that we display the interaction actions based on the current situation and relationship of the characters. For example, if the characters are friends, we do not show "fighting with each other". Another example is that the two characters are friends and they are close to each other, "approaching" and "fighting" are not displayed. The characters constitute a dynamic scene while they interact with

each other. The users can switch the scenes of a story smoothly while the story is proceeded to different stages.

3.3 Perspective switching

Our storytelling system has two view modes to watch the animated content of a story. In each view mode, there is a camera which is controlled to view the virtual environment at a certain direction. The first view mode is the storytelling view which shows the virtual environment for storytelling. At the beginning, the camera is set to look down on the virtual environment at a 45-degree angle, as the 3rd person perspective. However, the camera can be set to the eye position of a story character, as the 1st person perspective. The second view mode is the top-down view that is a floating minimap. The minimap enables parents to know the locations and statuses of characters and objects in a scene. The parents can use the two view modes to achieve certain impacts on listeners. The minimap supports the parents to perform perspective-switching actions while the storytelling view allows children to understand the progress of the story, such as watching the interaction of the story characters by using the 3rd person perspective or taking the role of a story character by using the 1st person perspective. We implement the system so that the camera transition between the 1st person and 3rd person perspective can be understandable for children. For example, if the parents want to switch from the 3rd person perspective to the 1st perspective of character A, the user can click on the camera on the minimap, and then click on character A. In the minimap (top-down field of view), a tangible arc will be formed between the camera and character A, and then the camera will move towards character A along the arc. After that, the camera movement will be dynamically displayed in the storytelling view. Once the camera moves to the position of character A, the first-person perspective of character A is adopted. In this way, the system gives an illusion to the users that they play the role of character A. Parents can set the speed of the camera transition process and explain to the children the educational significance of perspective switching.

3.4 Interactive storytelling

In the process of story preparation, users can use the edit interface to create, edit, or delete story scenes. Figure 2(a) shows the story preparation stage. All the characters and objects can be replicated and deleted in the virtual environment. The system also has offered some stories templates, whose backgrounds are based on fairytales, which can be edited by parents for making their family stories. Of course, the parents can create their own stories from scratch or may need to design their initial story scenes with their child. For example, they can populate a tree model at different locations and each tree has its own attributes in a story scene. The attributes of each character or object, such as location, size, orientation, and 3D model, are saved accordingly in a scene. Furthermore, the scenes or character objects can be reused in other stories, and they can achieve a new comprehensive story in a short time. Thus, they can construct a new story based on the materials of existing stories, thereby shortening the story preparation cycle.

During storytelling, the users can interactively manipulate the characters via the storytelling interface. Figure 2(b) shows the interactive storytelling stage. First, the users load the scene of the story. Then the users can start to narrate the story. Meanwhile, the users can control the characters so that the dynamic visual effects of the characters match to the story content. Thus, the users can see the characters interact with each other naturally. Narrating the story is a shared activity between parent and their children. The children will experience "their own adventure" or "someone's journey". Therefore, the storyteller can make the story characters come alive and storytelling be creative, lively, educational, and interesting.

The system allows the users to switch between the 1st person and 3rd person perspectives when viewing the characters and scenes during story telling. These two view perspectives are essential in storytelling, which give the users different feelings about the story characters. For example, when the 1st person perspective is applied to a character (e.g., a goat in Figure 1), the users can imagine that they play the role of the character. Thus, when the character makes a decision (e.g., yield way to another goat), the users can imagine that they make the same decision. Then the users can realize well the consequences of such decision. When using the 3rd person perspective, the users are the observers of the story characters. In this case, the users can judge well the overall behaviors of the story characters. The storyteller can bring the listeners to go through the entire story with different kinds of feelings when a person perspective is employed. For example, a goat is very small and another goat is large. When the 1st person perspective is applied, the feelings may not be the same on these two goats. The large goat may think that another goat is so small and would like to yield way to it. The small goat may think that the large goat looks fierce and then run away.

The following gives another example about the employment of the 1st person and 3rd person perspectives. This is an education story, *fools see an elephant*, which is a story about a metaphor for observing and judging things for generalization. Several fools see the body of the elephant, and they think that the part they see is the whole image of the elephant. Parents can use fool#1's perspective to see the nose of the elephant and tell their child that fool#1 thinks that the elephant is like a tube. Then the parents switch to fool#2's perspective to see the ear of the elephant and tell their child that fool#2 think that the elephant is like a fan. Finally, the parents employ the 3rd person perspective to view the entire elephant. The parents can teach their children not to be too subjective in their judgment of things, otherwise it is easy to be self-righteous. Perspectives play an important role in communication and empathy education.

4 CONCLUSIONS

In this paper, we present the design and implementation of a system for object-oriented 3D storytelling, which addresses the issues about the reusability and modifiability of the story contents. Family members can work in the interactive educational media space in which they can manipulate story characters and narrate via different person perspectives to collaboratively perform storytelling. This research provides a development direction of the storytelling system that is edutainment, free expression, and creativity. In the

future, we will plan to enhance the functions of the storytelling system. Then we will conduct a user study to analyze the behaviors of the users on operating the system. We will investigate how the users will perform in interaction, communication, and creation of collaborative stories. Furthermore, we would like to evaluate the role of joint attention for parent-child interaction. We believe that users' perspectives can be used as a tool of the interactive interface and make users be more empathetic with one another.

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